

## Multiple Choice (10 marks)

1. According to the recommendations for the nutritional management of patients with diabetes, the consumption of saturated fat should be:
  - (a) <10% of total daily energy
  - (b) <12%
  - (c) <15%
  - (d) <16%
2. Which of the following statements is true:
  - (a) Childhood stunting before 24 months of age affects short term cognitive performance
  - (b) Childhood stunting before 24 months of age affects long term cognitive performance
  - (c) Neither of the statements are correct
  - (d) Both of the statements are correct
3. The "fear of becoming fat" observed in patients with AN is best understood as a:
  - (a) Paranoid delusion
  - (b) Somatic delusion
  - (c) Overvalued idea
  - (d) Obsessive idea
4. Which of the following is the most frequently used anthropometric measure to estimate body mass?
  - (a) Fat-free mass
  - (b) Body fat
  - (c) Body height
  - (d) Body weight
5. Which of the following statements is not true?
  - (a) Docosahexaenoic acid can be synthesised from linolenic acid in vegans
  - (b) Docosahexaenoic acid is absent from vegan diets
  - (c) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment
  - (d) Microalgae can be used to synthesise docosahexaenoic acid

## True / False (10 marks)

*Answer True or False*

6. True or False: Facial flushing is seen in about 50% of East Asians due to a deficiency of ALDH2.
7. True or False: Intrauterine growth retardation (IUGR) is defined as birth weight below a population mean.
8. True or False: Orlistat is an approved drug for the treatment of obesity in Europe.
9. True or False: Sodium is a dietary component that has been proven to improve hypertension in humans.
10. True or False: Epigenetic changes can include DNA methylation and modify gene expression without changing the DNA sequence.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss Engel's law in food economics and explain how changes in income levels affect the proportion of income individuals spend on food. What implications does this have for health and nutrition?
12. Discuss the process by which  $\beta$ -carotene is converted into retinol in the body. How much retinol is typically produced from one mole of  $\beta$ -carotene, and why is this conversion important for vision?

*End of Paper*